Inventory Analytics: Workflow Documentation

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Project: Skyrovix Virtual Internship, Task 3 **Tool:** Power BI Desktop **Dataset:** clean_sales_data.csv **Due date:** 15 Nov 2026 **Student:** Karan Tormal

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1. Objective

Build an inventory analytics dashboard in Power BI to track stock levels, stockouts, fill rate and lost sales, and turn the findings into actions.

2. Dataset

Item	Detail
File	clean_sales_data.csv
Rows	76,000 (5 stores × 20 products × 760 days)
Period	1 Jan 2022 to 30 Jan 2024
Stores	S001 to S005
Products	P0001 to P0020
Key columns	Date, Store ID, Product ID, Category, Region, Inventory Level, Units Sold, Units Ordered, Demand, year, month

3. Data Preparation (Power Query)

1. Loaded the CSV file as the `clean_sales_data` table.
2. Set `Date` to Date type, using English (United States) locale.
3. Set numeric columns (Inventory Level, Units Sold, Units Ordered, Demand) to Whole Number.
4. Added `Month Number` using `Date.Month([Date])`.
5. Sorted `month` by `Month Number`, so charts run January to December.
6. Quality checks:
    - Null values: 0
    - Duplicate rows: 0
    - Rows where Units Sold exceeds Inventory Level: 0

4. KPI Definitions (DAX)

KPI	Definition	DAX logic
Stock Gap	Demand not met by sales	`SUM(Demand) – SUM(Units Sold)`
Stockout Rate	Share of rows with zero stock	`DIVIDE(COUNTROWS(FILTER(…, Inventory Level = 0)), COUNTROWS(…))`
Fill Rate	Share of demand that was sold	`DIVIDE(SUM(Units Sold), SUM(Demand))`
Days of Inventory	Stock cover in days	`DIVIDE(AVERAGE(Inventory Level), AVERAGE(Units Sold))`
Low Stock Rows	Rows with stock below 100 units	`COUNTROWS(FILTER(…, Inventory Level < 100))`
Sold Over Stock Rows	Rows where sales exceed stock (should be 0)	`COUNTROWS(FILTER(…, Units Sold > Inventory Level))`

5. Filters (Slicers)

- Date (range slider)
- Store ID
- Region
- Category

6. Dashboard Visuals

- **KPI cards:** Stockout Rate, Days of Inventory, Stock Gap, Low Stock Rows, Fill Rate
- **Line chart:** Monthly Demand vs Units Sold (X-axis: month, sorted by Month Number)
- **Bar chart:** Stockout Rate by Category (sorted descending)
- **Text box:** Actionable insights

7. Actionable Insights

#	Insight	Evidence	Action	KPI to track
1	Demand gap is largest in January	153,604 lost units in January, the largest gap	Raise stock 2 to 3 weeks before January. Plan purchase orders earlier for that month.	Monthly lost sales, January Fill Rate
2	Groceries runs out of stock most often	Stockout Rate 0.68%, highest of all categories	Set a higher reorder level for Groceries. Reorder before stock falls below 100 units.	Groceries Stockout Rate, Low Stock Rows
3	P0008 has stock but still misses demand	Fill Rate 81.5%, but Stockout Rate only 0.13%	Do not just add stock. Review P0008's pricing and demand forecast.	P0008 Fill Rate, price vs competitor pricing
4	Clothing has the lowest fill rate	Fill Rate 84.0%, lowest of all categories	Review the Clothing forecast and replenishment frequency. Consider promotions in slow periods.	Clothing Fill Rate
5	Stores S005 and S003 slightly underperform	Fill Rate 84.8% and 85.0%, below the 85.15% overall	Check replenishment timing for these stores. Compare order quantities with other stores.	Store Fill Rate, Days of Inventory

Note: Differences between stores and categories are small. Treat insights 4 and 5 as areas to monitor. Insights 1, 2 and 3 are the strongest findings.

8. Testing

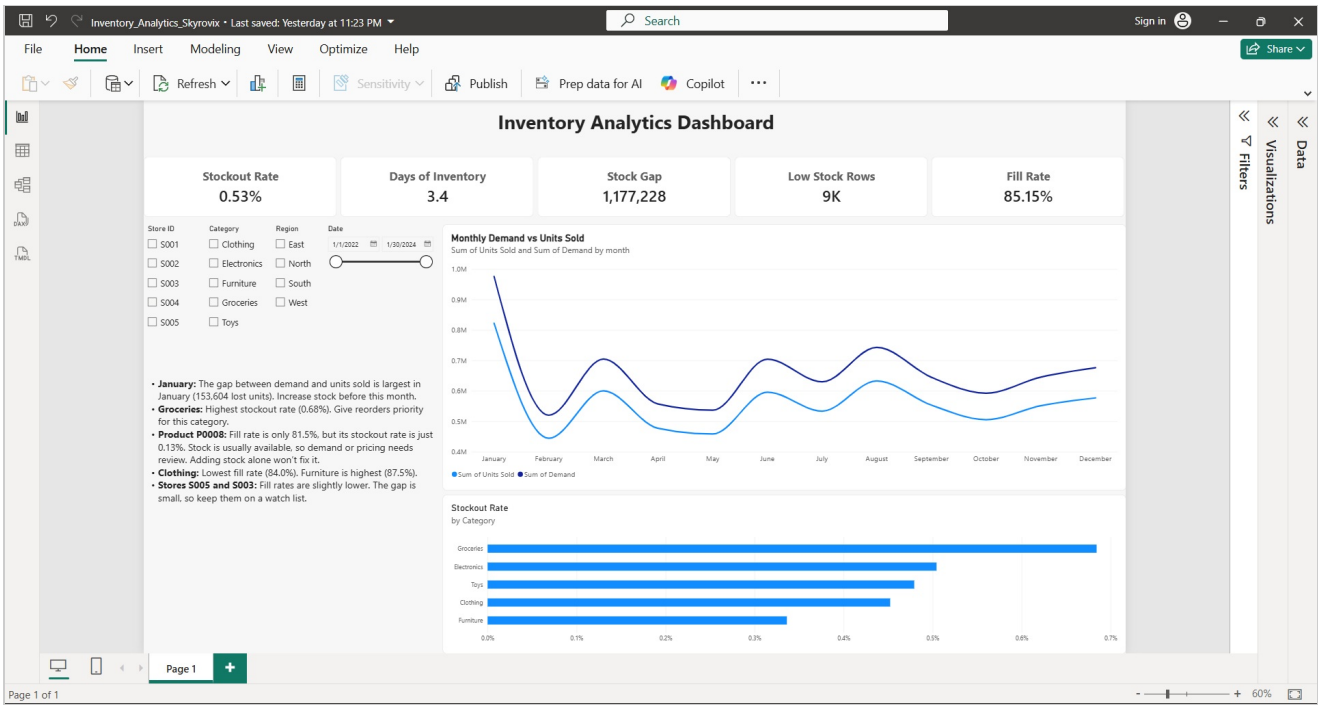
Dashboard values were checked against independent calculations in Python (pandas) on the same dataset.

#	Test	Expected	Result
1	Stock Gap, Store S001 selected	229,439	Pass
2	Stock Gap, no filter (all data)	1,177,228	Pass
3	Stockout Rate, Groceries selected	0.68%	Pass
4	Stock Gap, Groceries selected	550,349	Pass
5	Fill Rate, all data	85.15%	Pass
6	Stockout Rate, all data	0.53% (406 rows)	Pass
7	Days of Inventory, all data	3.4 (3.39 in pandas)	Pass
8	Sold Over Stock Rows, all data	0	Pass

9. Assumptions and Limitations

- **Stockout** is defined as Inventory Level = 0, because the dataset has no stockout flag.
- **Low stock threshold** of 100 units is an assumption, because the dataset has no Reorder Level column.
- **Category** changes for the same product across rows. Category is treated as a row-level attribute, not a fixed product property.
- **2024 data** covers only 1 to 30 January, so the 2024 trend is incomplete.
- **Currency** is not specified in the dataset, so no currency values are shown.

10. Dashboard Screenshot



Inventory Analytics Dashboard

11. Files Submitted

- `Inventory_Analytics_Skyrovix.pbix`: Power BI dashboard file
- `Dashboard_Screenshot.png`: Dashboard screenshot
- `Inventory_Analytics_Documentation.md`: This document